Scaled Dot Product Self Attention

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↳ 3 tokens, $d_k = d_v = 2$, w/ & w/o a causal mask

$$\text{Attention}(Q,K,V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}} + M\right)V$$

M is the mask. ($M=0 \rightarrow$ unmasked attention)

↳ Inputs

↳ each row is one token (token 1, token 2, token 3)

$$Q = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}, \quad K = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}, \quad V = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}$$

$Q, K, V \in \mathbb{R}^{3\times 2}$, $d_k = 2$, $d_v = 2$

↳ Transpose K & compute QK^T

$$K^T = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix} \in \mathbb{R}^{2\times 3}$$

↳ each entry is the dot product of row i of Q w/ row j of K:

$s_{11} = (1)(1) + (0)(0) = 1$ $\quad s_{12} = (1)(0) + (0)(1) = 0$

$s_{21} = (0)(1) + (1)(0) = 0$ $\quad s_{22} = (0)(0) + (1)(1) = 1$

$s_{31} = (1)(1) + (1)(0) = 1$ $\quad s_{32} = (1)(0) + (1)(1) = 1$

$s_{13} = (1)(1) + (0)(1) = 1$

$s_{23} = (0)(1) + (1)(1) = 1$

$s_{33} = (1)(1) + (1)(1) = 2$

$$QK^T = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 2 \end{bmatrix} \in \mathbb{R}^{3\times 3}$$

↳ Scale by $\sqrt{d_k}$

$$\sqrt{d_k} = \sqrt{2} = 1.4142$$

$$S = \frac{QK^T}{\sqrt{2}} = \begin{bmatrix} 1/\sqrt{2} & 0 & 1/\sqrt{2} \\ 0 & 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix} = \begin{bmatrix} 0.7071 & 0 & 0.7071 \\ 0 & 0.7071 & 0.7071 \\ 0.7071 & 0.7071 & 1.4142 \end{bmatrix}$$

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↳ Row wise softmax (no mask)

$$\text{softmax}(s_{ij}) = \frac{e^{s_{ij}}}{\sum_k e^{s_{ik}}} \qquad e^0 = 1.0,\ e^{0.7011} = 2.0281,\ e^{1.4142} = 4.1133$$

row 1 ↳ $[e^{0.7011}, e^0, e^{0.7011}] = [2.0281, 1.00, 2.0281]$, $\Sigma = 5.0562$

$$\left[\frac{2.0281}{5.0562}, \frac{1.000}{5.0562}, \frac{2.0281}{5.0562}\right] = [0.4011, 0.1978, 0.4011]$$

row 2 ↳ $[e^0, e^{0.7011}, e^{0.7011}] = [1.000, 2.0281, 2.0281]$, $\Sigma = 5.0562$

$$= [0.1978, 0.4011, 0.4011]$$

row 3 ↳ $[e^{0.7011}, e^{0.7011}, e^{1.4142}] = [2.0281, 2.0281, 4.1133]$, $\Sigma = 8.1695$

$$= [0.2483, 0.2483, 0.5035]$$

$$A = \text{softmax}(S) = \begin{bmatrix} 0.4011 & 0.1978 & 0.4011 \\ 0.1978 & 0.4011 & 0.4011 \\ 0.2483 & 0.2483 & 0.5035 \end{bmatrix}$$

↳ weighted sum of values: AV

↳ row 1: $0.4011(1) + 0.1978(3) + 0.4011(5) = 0.4011 + 0.5934 + 2.0055 = 3.00$

$0.4011(2) + 0.1978(4) + 0.4011(6) = 0.8022 + 0.7912 + 2.4066 = 4.00$

↳ row 2:

$0.1978(1) + 0.4011(3) + 0.4011(5) = 0.1978 + 1.2033 + 2.0055 = 3.407$

$0.1978(2) + 0.4011(4) + 0.4011(6) = 0.3956 + 1.6044 + 2.4066 = 4.407$

↳ row 3

$0.2483(1) + 0.2483(3) + 0.5035(5) = 0.2483 + 0.7449 + 2.5175 = 3.511$

$0.2483(2) + 0.2483(4) + 0.5035(6) = 0.4966 + 0.9932 + 3.021 = 4.511$

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$$\text{Output} = AV = \begin{bmatrix} 3.00 & 4.00 \\ 3.407 & 4.407 \\ 3.511 & 4.511 \end{bmatrix}$$

↳ causal mask

token i may only attend to tokens $j \le i$. Scores for future positions ($j > i$) are set to $-\infty$:

$$M = \begin{bmatrix} 0 & -\infty & -\infty \\ 0 & 0 & -\infty \\ 0 & 0 & 0 \end{bmatrix}, \quad S+M = \begin{bmatrix} 0.7071 & -\infty & -\infty \\ 0 & 0.7071 & -\infty \\ 0.7071 & 0.7071 & 1.4142 \end{bmatrix}$$

↳ softmax again, using $e^{-\infty} = 0$

row 1: $[2.0281, 0, 0]$, $\Sigma = 2.0281 \Rightarrow [1, 0, 0]$

row 2: $[1.00, 2.0281, 0]$, $\Sigma = 3.0281 \Rightarrow \left[\frac{1}{3.0281}, \frac{2.0281}{3.0281}, 0\right]$

row 3: no $-\infty$ entries $= [0.3302, 0.2483, 0.5035]$ (unchaged)

$$A_{masked} = \begin{bmatrix} 1 & 0 & 0 \\ 0.3302 & 0.6698 & 0 \\ 0.2483 & 0.2483 & 0.5035 \end{bmatrix}$$

↳ Masked output $A_{masked} V$

row 1: $1.[1,2] + 0.[3,4] + 0.[5,6] = [1.00, 2.00]$

row 2: $0.3302(1) + 0.6698(3) = 0.3302 + 2.0094 = 2.340$
$0.3302(2) + 0.6698(4) = 0.6604 + 2.6792 = 3.340$

row 3: unchanged $= [3.511, 4.511]$

$$\text{Masked output} = \begin{bmatrix} 1.000 & 2.000 \\ 2.340 & 3.340 \\ 3.511 & 4.511 \end{bmatrix}$$

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Token	No Mask	Causal Mask	Tokens it can see
Token 1	[3.000, 4.000]	[1.000, 2.000]	only token 1 (output = v_1)
Token 2	[3.407, 4.407]	[2.340, 3.340]	tokens 1 & 2
Token 3	[3.511, 4.511]	[3.511, 4.511]	tokens 1, 2 & 3 (unchanged)

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